EEG Data-Based Analysis of Paroxysmal Slow Wave Events Patterns in Brain Pathologies

Sol Amara Ganon[1], Alon Friedman[2] and Yaniv Zigel[1], *Senior Member, IEEE*

[1]Department of Biomedical Engineering, Ben-Gurion University of the Negev, Beer-Sheva, Israel

[2]Department of Brain and Cognitive Sciences, Ben-Gurion University of the Negev, Beer-Sheva, Israel and Dalhousie University, Halifax, Canada

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(1) Abstract

Recent studies have identified a distinct pattern of transient paroxysmal slow wave events (PSWEs) in patients with epilepsy and Alzheimer's disease, reflecting abnormal network dynamics [1],[2]. In this research, we aim to:

(1) Characterize the temporal and spatial characteristics of PSWEs in patients with epilepsy.

(2) Identify PSWEs features that will assist in the diagnosis of epilepsy, specifically drug-resistant epilepsy.

(3) Identify the sensitivity and specificity of selected combination of features that will help in differentiating between patients with epilepsy and non-epilepsy patients.

(2) Introduction

- **Epilepsy** is one of the most common neurological disorders characterized by the occurrence of spontaneous seizures [3].
- EEG is the primary diagnostic tool, but seizures often go undetected in short recordings, increasing the risk of misdiagnosis.
- Predicting epilepsy after a first seizure, brain injury or stroke remains a major clinical challenge.

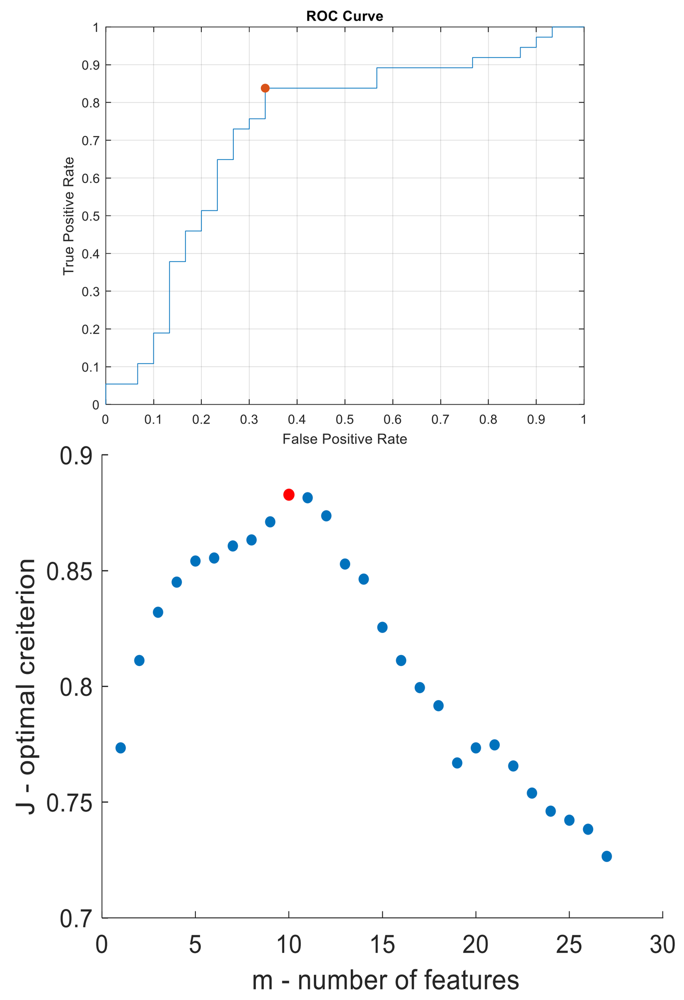
There is a pressing need for biomarkers that can aid in predicting and diagnosing epilepsy.

Paroxysmal Slow Wave Events (PSWEs), have emerged as promising EEG biomarkers for various brain pathologies. PSWEs are transient slow-wave events marked by a network shift from normal activity to brief periods of low-frequency activity. These events occur more frequently and exhibit longer durations in individuals with epilepsy compared to those who are seizure-free [2],[3].

PSWE defined when: Median power frequency (MPF) < 6 [Hz] for at least 5 [s].

(3) Methods

EEG recording
↓
Pre-processing
BPF 1-45 [Hz] | DC removal | ICA
↓
PSWE detection
↓
Feature extraction
↓
Feature selection
Correlation-Based filter approach | Forward feature selection
↓
Classification model



Num.	Features group	Example
	Time domain	
1-11	Amplitude distribution	Energy: $E = \frac{1}{N}\sum_{n=1}^{N} x_n^2$
12-15	non-linear energy	$ne(i) = data(i)^2 - data(i-1)\cdot data(i+1)$ Mean(ne)
	Frequency domain	
16-20	FFT features	STD FFT
21-26	Relative power in brain waves	$RP = \frac{1}{FT}\cdot\sum_{j=a}^{b} f_j\,;FT = \sum_{\text{all freq}} f_j$ Relative power (RP) in Delta 1–3 Hz
27-39	Other features per patients	Number of PSWEs

(4) Results

The TUH EEG Epilepsy Corpus dataset [4]:

188 Epilepsy subjects & 161 subjects without a known diagnosis of epilepsy

The University Hospital Bonn dataset:

62 drug-resistant subjects & 70 drug-responsive subjects (no seizure for at least one year with anti-seizure medication)

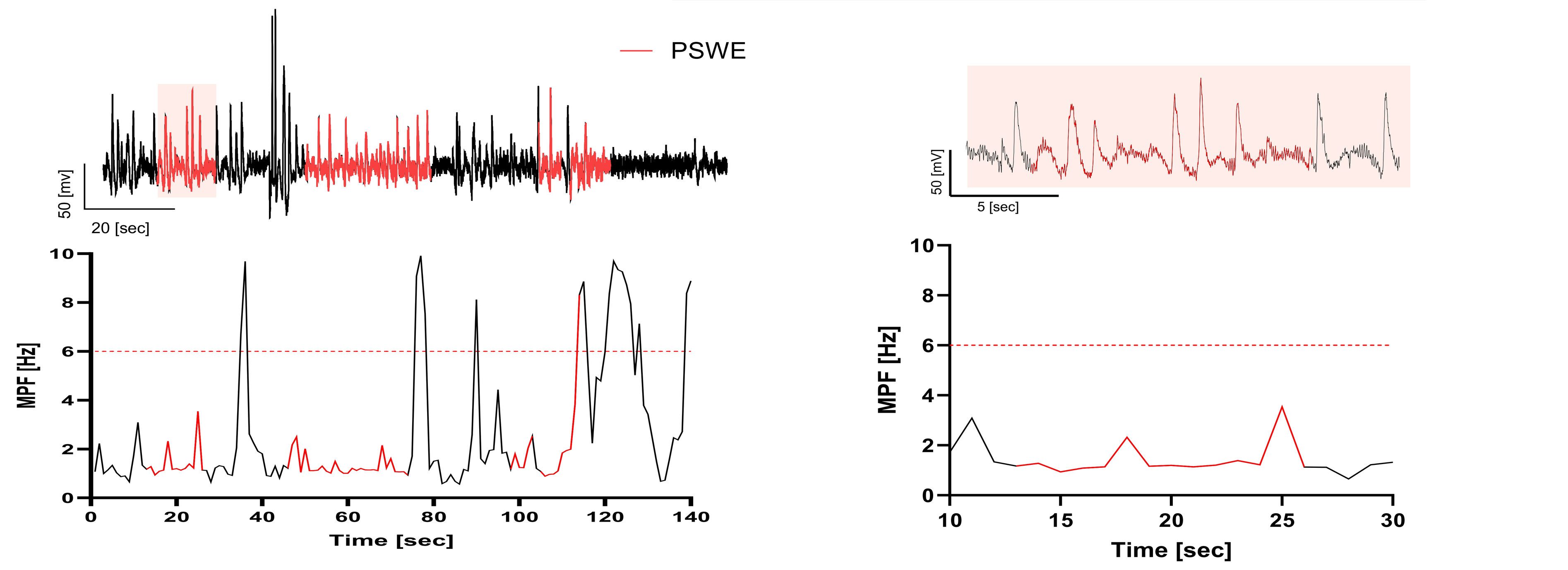


Fig 1. PSWE detection. Left: EEG signal from a single electrode and the corresponding median power frequency (MPF) per second graph. Right: Zoomed-in view of the first detected PSWE event.

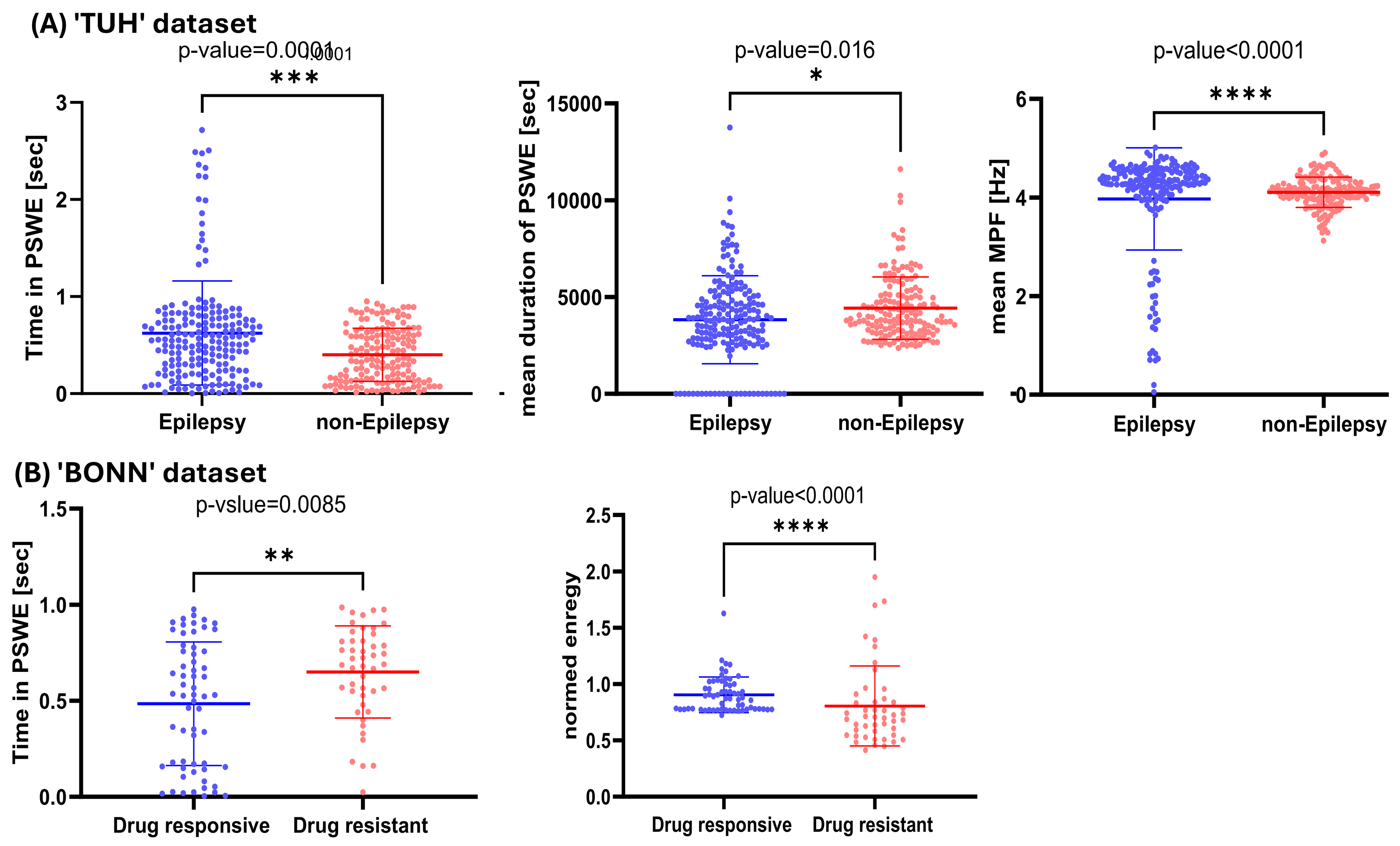


Fig 2. Examples of Significant differences of PSWE in Patient Groups.

(A) Gaussian SVM

Sensitivity	Specificity	Accuracy
78.12%	78.38%	78.26%

(B) Logistic Regression

Sensitivity	Specificity	Accuracy
88.89%	93.33%	91.67%

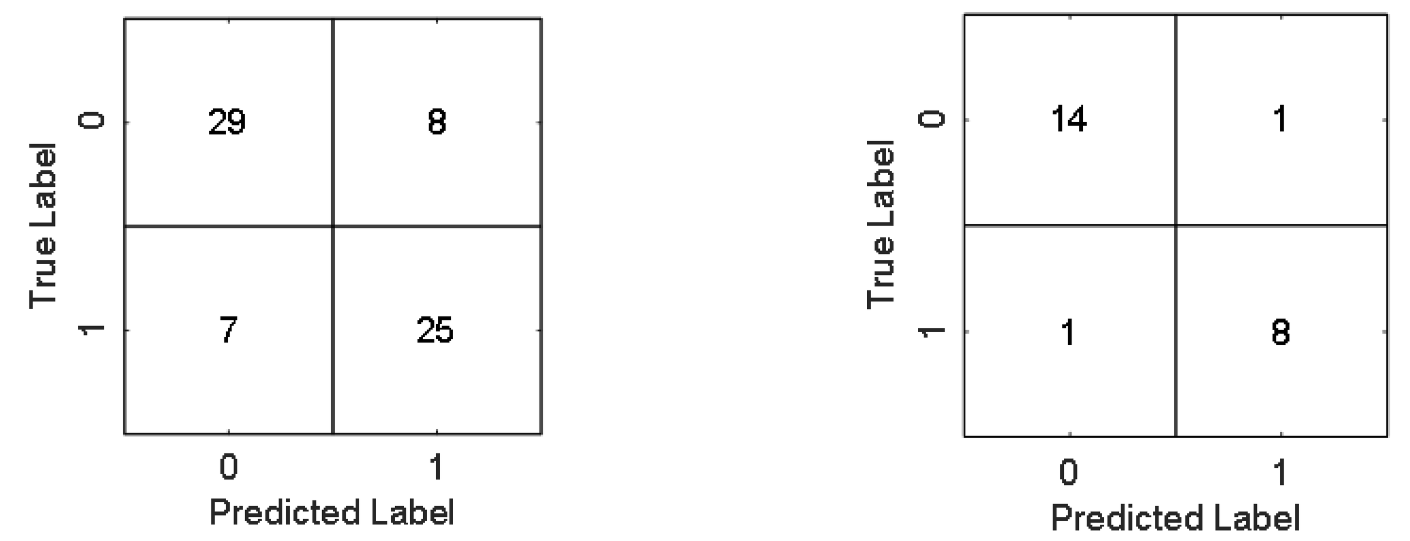


Fig 3. Classification models performance. (A) TUH dataset: 1 – Epilepsy, 0 – non-Epilepsy; (B) BONN dataset: 1 – Drug resistant, 0 – Drug responsive.

(5) Discussion & Conclusions

- We present a novel approach for diagnosing epilepsy by analyzing the features of PSWEs.
- PSWEs can serve as reliable biomarkers for distinguishing epilepsy patients from healthy individuals, as well as for the diagnosis of drug-resistant patients.
- PSWE-based models may aid clinical decision-making and treatment planning.

References

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